

CURRICULUM VITAE (CV)

Jin-Bin Im - Ph.D. Candidate (expected in August 2026)

Hanyang University, Department of Architectural Engineering

Client Briefing & Program Management (CB&PM) Lab

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SUMMARY

- **Expertise** – AI-driven human-centered construction intelligence, smart sensing and real-time monitoring, human behavioral data analysis, BIM-integrated digital workflows
- **Publication** – I have published multiple peer-reviewed articles in high-impact international journals in the fields of construction engineering, digital construction, and the built environment, including *Building and Environment* (IF 7.6), *Journal of Management in Engineering* (IF 7.0), and *Architectural Science Review* (A&HCI, top 2.58%).
- **Academic Service** – I have served the academic communities as a reviewer of high impact journals, such as, *Building and Environment*, *Journal of Management in Engineering*, and *Architectural Engineering and Design Management*.

RESEARCH INTERESTS

- **Smart Construction Intelligence & Monitoring**
 - AI pipelines for real-time construction site monitoring and automated quality management
 - Data-driven decision support systems integrating on-site sensing with digital construction workflows
 - Personalized and adaptive learning systems for construction site operators
- **Human-Centered Digital Workflows & BIM Integration**
 - End-to-end pipelines from multimodal data acquisition to BIM-linked structured output
 - Natural language processing for translating user requirements into design parameters
 - Interoperable digital construction frameworks supporting downstream lifecycle analysis
- **Immersive Virtual Environment for Smart Sensing & Assessment**
 - XR (VR/MR) platforms for real-time physiological and cognitive performance monitoring
 - Human-machine interaction in complex, dynamic spatial environments
 - Sensor-integrated immersive environments for health and well-being assessment

EDUCATION

Hanyang University

Mar. 2020–Aug. 2026

M.S/Ph.D. in Architectural Engineering, Construction Management

Dissertation title: “Narrative Feedback-Driven Affective Embedding Space Learning and Optimal Transport based Automated Spatial Design Revision”

Hanyang University

Mar. 2014–Feb. 2020

B.S. in Interior Architecture Design

EXPERIENCE

- Research Assistant**, Hanyang University, *Dept. of Architectural Engineering* Mar. 2020–present
- *Development of Artificial Intelligence-based Real-Time Construction Quality Management System.* April. 2026–present
 - *Development of Artificial Intelligence-based Real-Time Construction Quality Management System.* May. 2024–present
 - *Study on Quality Control of Educational Facilities using Virtual Environments.* Oct. 2023–Oct. 2024
 - *Development of Decision-making Framework to Support Personalized Spatial Composition based on Temperament, Perception and Bio-markers Applying Digital Twin.* Sep. 2021–Feb. 2024
 - *Construction Quality Control using Digital Twin Technology.* Apr. 2021–Dec. 2022
 - *Establishment of Standard Guidelines for Daycare Centers in Police Stations.* Jun–Sep. 2021
- Internship**, SIFC Property Korea YH Nov. 2019–May. 2020
- *Property management for IFC Seoul, particularly an IFC Mall (85,444m² GFA).*
- Research Intern**, Hanyang University, *Dept. of Interior Architecture Design* May–Jul. 2019
- *Spatial Preference Modeling based on Immersive Virtual Environment and Choice Modeling Theory.*

TEACHING

- Lecturer**, Korea University of Technology and Education 2026
- *AI-based Optimization Technology for Architectural Design*
- Teaching Fellow**, Hanyang University, Department of Architectural Engineering 2024–2025
- *PBL4004 Architectural Engineering Capstone PBL* Fall 2025
 - *ENE4044 Computer Engineering* Spring 2024, 2025

SERVICE

- Reviewer**, *23rd Smart and Sustainable Built Environment (SASBE)* 2026
- Reviewer**, *Results in Engineering* 2026–present
- Reviewer**, *Journal of Management in Engineering* 2026–present
- Reviewer**, *Architectural Engineering and Design Management* 2025–present
- Reviewer**, *Building and Environment* 2025–present

PATENT

- *South Korea Patent, No. 10-2421360, “System for Generating 3D LiDAR Scanning Position.”* Jul. 2022

• REFERRED JOURNAL

- [J13] Hong, R. L., Yoon, J. H., **Im, J. B.**, & Kim, J. H.* (in press). Prediction of Actual Safety and Health Management Expenses for Water Supply System Construction Projects in Korea through Machine Learning. *Journal of Civil Engineering and Management*, Vilnius Tech Press. [SCIE, Q1, 14.05%, IF 3.7]
- [J12] Lee, K. T., Shin, S. H., **Im, J. B.**, Choi, H., & Kim, J. H.* (in press). Integrating AI-Enhanced and Conventional Market Selection Approaches for the Construction Industry: Insights from Indian Firms. *Journal of Civil Engineering and Management*, Vilnius Tech Press. [SCIE, Q1, 14.05%, IF 3.7]
- [J11] Lee, K. T., **Im, J. B.**, Hong, R. L., & Kim, J. H.* (in press). Enhancing Retrieval-Augmented Generation Performance through Network Analysis of Question Types in Public Procurement Law. *Journal of Management in Engineering*, ASCE. [SCIE, Q1, 4.35%, IF 7.0] <https://doi.org/10.1061/JMENE/MEENG-7429>
- [J10] **Im, J. B.**, Hong, R. L., Joo, M., Zhang, E., & Kim, J. H.* (2025). Visual Fidelity Effects on Occupants' Performance, Mental States, and Emotions in Mixed Reality Environments. *Architectural Science Review*, Taylor & Francis, 1–21. [A&HCI, Q1, 2.58%, IF 1.8] <https://doi.org/10.1080/00038628.2025.2584180>
- [J9] Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2025). Effects of Human Perception Dimensions Constituting Individual Temperament and Experience on Indoor Space Composition Decision-making Presented in a Mixed-reality Environment. *Architectural Science Review*, Taylor & Francis, 1–16. [A&HCI, Q1, 2.58%, IF 1.8] <https://doi.org/10.1080/00038628.2025.2510300>
- [J8] Joo, M., **Im, J. B.**, Lee, K. T., & Kim, J. H.* (2025). Comparative Analysis of Work Performance and Physiological Response Based on Office Types and Indoor Space Design. *Architectural Engineering and Design Management*, Taylor & Francis, 1–17. [SCIE, Q2, 34.3%, IF 2.7] <https://doi.org/10.1080/17452007.2025.2490958>
- [J7] **Im, J. B.**, Joo, M., Lee, K. T., & Kim, J. H.* (2025). A mixed reality approach to identify cognitive performance and mental states in preferred vs. non-preferred individual and collaborative work environments. *Building and Environment*, Elsevier, 274, 112806. [SCIE, Q1, 4.1%, IF 7.1] <https://doi.org/10.1016/j.buildenv.2025.112806>
- [J6] Park, S. J., **Im, J. B.**, Yoon, H. S., & Kim, J. H.* (2025). Data Augmentation Approaches for Estimating Curtain Wall Construction Duration in High-Rise Buildings. *Buildings*, MDPI, 15(4), 583. [SCIE, Q2, 27.57%, IF 3.1] <https://doi.org/10.3390/buildings15040583>
- [J5] Kim, H. J., Kim, J. H.*, & **Im, J. B.*** (2023). Forecasting offline retail sales in the COVID-19 pandemic period: A case study of a complex shopping mall in South Korea. *Buildings*, MDPI, 13(3), 627. [SCIE, Q2, 40.4%, IF 3.3] <https://doi.org/10.3390/buildings13030627>
- [J4] Lee, K. T., **Im, J. B.**, Park, S. J., & Kim, J. H.* (2023). Identifying improvements priorities in daycare centre space performance from teachers' perspectives: application of importance–performance analysis and structural equation modelling. *Engineering, Construction and Architectural Management*, Emerald, 32(1), 639–672. [SCIE, Q1, 22.3%, IF 3.6] <https://doi.org/10.1108/ECAM-08-2022-0777>
- [J3] Lee, K. T., **Im, J. B.**, Park, S. J., & Kim, J. H.* (2022). Conceptual framework to support personalized indoor space design decision-making: A systematic literature review. *Buildings*, MDPI, 12(6), 716. [SCIE, Q2, 44.1%, IF 2.6] <https://doi.org/10.3390/buildings12060716>
- [J2] Park, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2022). The need for smart architecture caused by the impact of COVID-19 upon architecture and city: A systematic literature review. *Sustainability*, MDPI, 14(13), 7900. [SCIE, Q2, 45.0%, IF 3.2] <https://doi.org/10.3390/su14137900>
- [J1] Nindartin, A., Moon, H. W., Park, S. J., Lee, K. T., **Im, J. B.***, & Kim, J. H.* (2022). Influencing of the Building Energy Policies upon the Efficiency of Energy Consumption: The Case of Courthouse Buildings in South Korea. *Energies*, MDPI, 15(18), 6679 [SCIE, Q3, 60.9%, IF 3.0] <https://doi.org/10.3390/en15186679>

- **REFERRED CONFERENCE**

- [C9] Hong, R. L., **Im, J. B.**, Shim, W. S., Wang, S. H., Ha, W. C., & Kim, J. H. * (2026). Automatic Semantic Parsing of Structural Drawings and Rebar Schedules Based on a Hybrid Deep Learning Framework, *the 43rd International Symposium of Automation and Robotics in Construction (ISARC 2026)*, Singapore, 22–26 June, 2026. (provisionally accepted)
- [C8] **Im, J. B.**, Hong, R. L., Choi, C. H., Ha, J. E., & Kim, J. H. * (2026). Optimal Transport-based Affective User Requirements Realignment for Automated Minimal Disruptive Spatial Design Revision in BIM Environment, *the 43rd International Symposium of Automation and Robotics in Construction (ISARC 2026)*, Singapore, 22–26 June, 2026. (provisionally accepted)
- [C7] **Im, J. B.**, Ye, S. J., Zhang, E., Lee, K. M., & Kim, J. H. * (2026). Assessing Material and Labor Cost Indices Impacts on Construction Firms' Financial Performance: A Machine Learning Comparison with Feature Engineering, *the 43rd International Symposium of Automation and Robotics in Construction (ISARC 2026)*, Singapore, 22–26 June, 2026. (provisionally accepted)
- [C6] Zhang, E., **Im, J. B.**, Xu, L., Park, S. Y., & Kim, J. H. * (2026). The Influence of Window Characteristics on Occupant Well-Being and Perception in Indoor Environments: A Review, *the 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings (IAQVEC 2026)*, Los Angeles, CA, 18–22 May, 2026. (accepted)
- [C5] Zhang, E., Choi, S. W., **Im, J. B.**, Baek, Y. G., Kim, J. H., & Lee, K. T. * (2025). Construction Conflict Resolution Strategies Focused on Two-level Game Theory: A Case Study Analysis, *the 11th International Conference on Construction Engineering and Project Management (ICCEPM 2025)*, pages 152–119, Jeju, South Korea, 17–20 November, 2025. <https://doi.org/10.6106/ICCEPM.2025.0152>
- [C4] Ochirsuren, N., Kim, G. T., **Im, J. B.**, Hong, R. L., Xu, L., & Kim, J. H. * (2025). A BPMN-based Framework for Real-time Rebar Arrangement Inspection on Construction Sites, *the 11th International Conference on Construction Engineering and Project Management (ICCEPM 2025)*, Pages 136–143, Jeju, South Korea, 17–20 November, 2025. <https://doi.org/10.6106/ICCEPM.2025.0136>
- [C3] Hong, R. L., **Im, J. B.**, Zhang, E., Park, S. J., Ochirsuren, N., Beak, Y. G., Kang, S. H., Wang, S., Lee, C. S., Cheong, S. W., & Kim, J. H. * (2025). Development of a Petrochemical Plant Pipe Counting Method Based on Improved Mask R-CNN and Transformer. *the 42nd International Symposium of Automation and Robotics in Construction (ISARC 2025)*, pages 972–979, Montreal, Canada, 28–31 July, 2025. <https://doi.org/10.22260/ISARC2025/0126>
- [C2] **Im, J. B.**, Hong, R.L., Jang, J. H., Zhang, E., Shin, S. H., Joo, M., Lee, K. W., Lee, K. M., Lee, K. H., Park, S. Y., & Kim, J. H. * (2025). Effects of Visual Fidelity on Task Performance and Mental Workload in IVEs. *the 42nd International Symposium of Automation and Robotics in Construction (ISARC 2025)*, pages 909–916, Montreal, Canada, 28–31 July, 2025. (Presenter) <https://doi.org/10.22260/ISARC2025/0118>
- [C1] Jang, J. H., **Im, J. B.***, Zhang, E. L., Joo, M. B., Kang, S. H., & Kim, J. H. * (2024). A Framework for Assessing the Learning Performance and Creativity in Spatial Features by Immersive Virtual Environments. *the 10th International Conference on Construction Engineering and Project Management (ICCEPM 2024)*, pages 621–628, Sapporo, Japan, 29 July – 1 August, 2024. (Presenter) <https://doi.org/10.6106/ICCEPM.2024.0621>

DOMESTIC PUBLICATION

- **REFERRED JOURNAL**

- [6] Shin, S. H., Lee, K. M., **Im, J. B.**, & Kim, J. H. * (2026). Factor Analysis on Cost Overrun by Introducing FMEA and Proposal for Improving Approximate Estimation Reliability in Apartment Projects, *Journal of the Architectural Institute of Korea*, 42(3), 339–347. [Scopus] <https://doi.org/10.5659/JAIK.2026.42.3.339>
- [5] Zhang E. L., Jo, S. H., **Im, J. B.**, & Kim, J. H. * (2024). Importance and Performance Analysis on Stakeholders Perceptions against Public Rental Housing Status and Directions to Improve Occupants Satisfaction. *Journal of the Architectural Institute of Korea*, 41(1), 301–312. [Scopus] <https://doi.org/10.5659/JAIK.2025.41.1.301>

- [4] Lee, K. Y., Ding, Z. S., **Im, J. B.**, Park, S. J., & Kim, J. H.* (2024). Investigation of Construction Duration Influence Factors and Comparison of Artificial Intelligence Methods for Estimating Appropriate Construction Duration in Public Sewerage Construction. *Korea Institute of Construction Engineering and Management*, 25(6), 66–74. [KCI] <https://dx.doi.org/10.6106/KJCEM.2024.25.6.066>
- [3] Kim, D. H., Ye, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2023). Developing a Comprehensive Framework for Detailed Facility Planning in Office Building Grade Assessment. *Journal of the Architectural Institute of Korea*, 39(9), 69–80. [Scopus] <https://doi.org/10.5659/JAIK.2023.39.9.69>
- [2] **Im, J. B.**, Hwang, S. J., Jang, J. H., Kim, J. H., & Lee, K. T.* (2022). Introduction of mixed reality on special tree landscape design review phase. *Journal of the Architectural Institute of Korea*, 38(8), 71–81. [Scopus] <https://doi.org/10.5659/JAIK.2022.38.8.71>
- [1] Kim, H. S., **Im, J. B.**, Lee, K. T., Kim, H. J., & Kim, J. H.* (2021). Proposals for Feasibility Study and Optimal Combination of Renewable Energy Generation Systems in Baengnyeongdo and Oedo using HOMER. *Journal of the Architectural Institute of Korea*, 37(12), 347–356. [Scopus] <https://doi.org/10.5659/JAIK.2021.37.12.347>

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- [5] Kim, D. H., Ye, S. J., **Im, J. B.**, & Kim, J. H.* (2023). Developing a Comprehensive framework for Detailed Facility Planning in Office Building Grade Assessment. *Spring Annual Conference of AIK*, pages 769–772.
- [4] **Im, J. B.**, Ding, Z. S., & Kim, J. H.* (2023). A Case Study of Complex Shopping Mall on Factors Relevant to COVID-19 Affecting Offline Retail Sales, *Spring Annual Conference of AIK*, pages 735–736.
- [3] Lee, K. T., **Im, J. B.**, Ding, Z. S., Hong, R. L., & Kim, J. H.* (2022). An Analysis of User’s Emotions and Task Performance in Indoor Space Implemented by Mixed-Reality, *Autumn Annual Conference of AIK*, pages 1051–1054.
- [2] Park, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2022). Design of Experiments for Voxel Down-sampling Indoor Point Cloud Data, *Autumn Annual Conference of AIK*, pages 1082–1085.
- [1] **Im, J. B.**, Lee, K. T., Kim, H. J., Nindartin, A., Nour, N., & Kim, J. H.* (2022). Forecasting of Offline Complex Shopping Mall Retail Sales after COVID-19 pandemic, *Autumn Annual Conference of AIK*, pages 1060–1061.

IN-PROGRESS RESEARCH

[In 2nd Round Revision] Prediction of structural frame works safety and health management expenses in apartment construction projects by introducing interpretable machine learning. Submitted to *Engineering, Construction, and Architectural Management*.

[In 1st Round Revision] Visual quality features in immersive virtual environments: A systematic review of their impact on user experience. Submitted to *Virtual Reality*.

[In Review] Optimizing Immersive Virtual Environments for Architectural Design Review: The Impact of Visualization Quality on Occupant-Centric Performance Assessment. Submitted to *Journal of Building Engineering*.

[In Review] Strategic Pathways for e-Methanol Deployment in the Low-Carbon Transition: Policy Modeling of Cost Competitiveness and Flexibility Requirements in Korea. Submitted to *Energy Strategy Reviews*.

[In Review] Beyond Viewing: How Installation Artwork Design Modality Shapes Visitor Satisfaction Through Affective and Behavioral Pathways. Submitted to *Scientific Reports*.

[In Review] Weather-adaptive automated pipe counting for open-air construction storage: Leveraging synthetic data augmentation and YOLO-Network Family. Submitted to *Developments in the Built Environment*.

[Manuscript Completed] Effect of Changing Sequence in Semi-Top-Down Construction Works to Optimize Time and Cost: A Case Study on Korean Apartment Projects.

[In Preparation] An Automated Framework for Minimally Disruptive Spatial Design Revision: Reflecting Affective User Requirements through Natural Language Processing and Optimal Transport.

[In Preparation] An Affective Engineering Framework for Modeling Emotional Coherence in Spatial Design.

REFERENCE

Prof. Ju-Hyung Kim, Dean of Academic Affairs, Professor, Hanyang University

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Prof. Joon Oh Seo, Associate Professor, Hanyang University

Relationship: Dissertation Committee Member and Collaborator

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Prof. Seunghyeon Wang, Lecturer, University College London

Relationship: Collaborator

E-mail: seung-hyun.wang@ucl.ac.uk

Prof. Kyung-Tae Lee, Assistant Professor, Kangwon National University

Relationship: Graduate School Colleague and Collaborator

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